

Supplementary of “Label Confidence Recovery with High-order Label Correlation in Partial Multi-label Learning”

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A Details of Baseline Methods

In this appendix, we provide detailed descriptions of all baseline methods used in our experiments. These methods encompass two major paradigms in PML: two-stage methods and unified methods.

A.1 Two-stage methods

These methods decompose the PML problem into sequential stages of label disambiguation followed by classifier training.

PARTICLE (PARTIAL multi-label learning via Credible Label Elicitation) [Zhang and Fang, 2021]. This method develops a probabilistic model that iteratively updates label confidences through expectation-maximization. It models the generation process of candidate labels and estimates the confidence of each candidate label being a ground-truth label.

PML-LRS (Partial Multi-label Learning by Low-Rank and Sparse decomposition) [Sun *et al.*, 2019]. This method utilizes low-rank and sparse decomposition to recover ground-truth label matrices from noisy candidate sets. It assumes that the ground-truth label matrix exhibits low-rank structure due to label correlations, while the noise introduced in candidate sets is sparse.

PML-NI (Partial Multi-label Learning with Noisy label Identification) [Xie and Huang, 2021]. This method explicitly identifies noisy labels via a noise identification module before training, which leverages both instance features and label correlations to detect false positive labels.

PML-MD (Partial Multi-Label Learning with Meta Disambiguation) [Xie *et al.*, 2021]. This method employs meta-learning techniques to adaptively estimate label confidences. The meta-learning framework enables the model to generalize disambiguation strategies learned from one subset of data to others.

A.2 Unified methods

These methods jointly optimize label disambiguation and classifier training to enable mutual reinforcement between the two components.

GLC (Global-Local label Correlation for partial multi-label learning) [Sun *et al.*, 2022]. This method integrates global and local label correlations through low-rank representation and label manifold regularization.

PAMB (PARTIAL Multi-label learning via BINARY decomposition) [Liu *et al.*, 2023]. This method enables binary decomposition for partial multi-label learning by decomposing the multi-label problem into multiple binary classification problems. It handles the ambiguity during decomposition and uses consistency constraints across binary classifiers to improve disambiguation quality.

PLAIN (deep Partial multi-Label model with grAph-disambIguationN) [Wang *et al.*, 2023]. This method constructs a label correlation graph that captures pairwise dependencies and uses graph neural networks to propagate information across labels.

PARD (Partial multi-label learning with probAbilistic-grAphical Disambiguation) [Hang and Zhang, 2023]. This method models the joint distribution of labels through graphical structures and performs inference to determine the most likely true label configurations.

WPML³CP (Wasserstein Partial Multi-Label Learning with dual Label Correlation Perspectives) [Li *et al.*, 2024]. This method captures label correlations from both measuring and modeling perspectives by employing Wasserstein distance.

PML-FSMIR (Partial Multi-label Learning via Feature Selection and Manifold-regularized Instance Ranking) [Gao *et al.*, 2025]. This is a recent method that combines feature selection with manifold regularization for partial multi-label learning. It performs feature selection to identify discriminative features while using manifold regularization to preserve the geometric structure of the data.

B Results on UCI Datasets ($m = \{100, 50\}$)

To comprehensively evaluate the robustness of REHC-PML across varying noise levels, we conduct additional experiments on UCI multi-label datasets with noise ratios $m \in \{50, 100, 150\}$.

Table 1 presents the experimental results under moderate noise conditions. REHC-PML maintains its performance advantage as noise levels decrease, achieving the best Average Precision on all eight datasets. Significant improvements are observed on *bibtex*, *bow-train*, and *nuswide*. The performance gains are especially pronounced on datasets with high label dimensionality such as *delicious*, where REHC-PML achieves 0.402 compared to 0.361 for the second-best PMLNI (+11.4%). For ranking-based metrics, REHC-PML consistently outperforms baselines across most datasets. On *enron*, PML-FSMIR

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Methods	bibtex	bookmarks	bow-train	computers	delicious	enron	entertainment	nuswide
Average Precision $\uparrow$								
REHC-PML	0.533$\pm$0.006	0.500$\pm$0.002	0.761$\pm$0.004	0.762$\pm$0.007	0.402$\pm$0.004	0.695$\pm$0.015	0.782$\pm$0.001	0.673$\pm$0.001
PML-FSMIR	0.501 $\pm$ 0.009	-	0.670 $\pm$ 0.004	-	0.267 $\pm$ 0.007	0.669 $\pm$ 0.014	-	-
WML3CP	0.425 $\pm$ 0.016	0.438 $\pm$ 0.002	0.670 $\pm$ 0.022	0.700 $\pm$ 0.010	0.260 $\pm$ 0.003	0.604 $\pm$ 0.004	0.694 $\pm$ 0.011	0.635 $\pm$ 0.002
PLAIN	0.296 $\pm$ 0.011	0.274 $\pm$ 0.003	0.705 $\pm$ 0.006	0.709 $\pm$ 0.003	0.222 $\pm$ 0.005	0.664 $\pm$ 0.016	0.720 $\pm$ 0.002	0.637 $\pm$ 0.001
PARD	0.524 $\pm$ 0.020	0.419 $\pm$ 0.004	0.677 $\pm$ 0.002	0.675 $\pm$ 0.009	0.285 $\pm$ 0.005	0.682 $\pm$ 0.010	0.695 $\pm$ 0.004	0.582 $\pm$ 0.002
PAMB	0.140 $\pm$ 0.011	-	0.504 $\pm$ 0.021	-	-	0.477 $\pm$ 0.073	-	-
GLC	0.282 $\pm$ 0.010	0.421 $\pm$ 0.003	0.652 $\pm$ 0.003	0.659 $\pm$ 0.004	0.050 $\pm$ 0.002	0.577 $\pm$ 0.010	0.611 $\pm$ 0.005	0.568 $\pm$ 0.002
PMLMD	0.501 $\pm$ 0.006	0.370 $\pm$ 0.009	0.678 $\pm$ 0.005	0.722 $\pm$ 0.006	0.339 $\pm$ 0.003	0.688 $\pm$ 0.013	0.728 $\pm$ 0.006	0.586 $\pm$ 0.011
PMLNI	0.537$\pm$0.010	0.470 $\pm$ 0.003	0.713 $\pm$ 0.004	0.666 $\pm$ 0.005	0.361 $\pm$ 0.002	0.539 $\pm$ 0.008	0.652 $\pm$ 0.008	0.637 $\pm$ 0.001
PARTICLE	0.263 $\pm$ 0.008	-	0.694 $\pm$ 0.015	-	-	0.593 $\pm$ 0.059	-	-
Coverage Error $\downarrow$								
REHC-PML	0.173 $\pm$ 0.011	0.147$\pm$0.003	0.171$\pm$0.003	0.119 $\pm$ 0.005	0.521$\pm$0.005	0.254 $\pm$ 0.011	0.131 $\pm$ 0.014	0.116$\pm$0.001
PML-FSMIR	0.193 $\pm$ 0.005	-	0.198 $\pm$ 0.003	-	0.639 $\pm$ 0.007	0.231$\pm$0.010	-	-
WML3CP	0.227 $\pm$ 0.008	0.149 $\pm$ 0.002	0.178 $\pm$ 0.009	0.128 $\pm$ 0.008	0.613 $\pm$ 0.006	0.324 $\pm$ 0.008	0.128 $\pm$ 0.016	0.135 $\pm$ 0.002
PLAIN	0.362 $\pm$ 0.020	0.317 $\pm$ 0.004	0.256 $\pm$ 0.005	0.155 $\pm$ 0.004	0.761 $\pm$ 0.011	0.271 $\pm$ 0.005	0.164 $\pm$ 0.005	0.172 $\pm$ 0.002
PARD	0.141$\pm$0.009	0.185 $\pm$ 0.016	0.203 $\pm$ 0.003	0.110$\pm$0.003	0.853 $\pm$ 0.025	0.251 $\pm$ 0.005	0.125$\pm$0.003	0.166 $\pm$ 0.002
PAMB	0.570 $\pm$ 0.016	-	0.559 $\pm$ 0.015	-	-	0.504 $\pm$ 0.084	-	-
GLC	0.241 $\pm$ 0.007	0.158 $\pm$ 0.002	0.205 $\pm$ 0.003	0.116 $\pm$ 0.003	0.955 $\pm$ 0.002	0.265 $\pm$ 0.012	0.140 $\pm$ 0.004	0.160 $\pm$ 0.001
PMLMD	0.182 $\pm$ 0.006	0.206 $\pm$ 0.004	0.249 $\pm$ 0.020	0.117 $\pm$ 0.006	0.571 $\pm$ 0.006	0.246 $\pm$ 0.010	0.127 $\pm$ 0.006	0.183 $\pm$ 0.004
PMLNI	0.235 $\pm$ 0.004	0.189 $\pm$ 0.002	0.194 $\pm$ 0.005	0.170 $\pm$ 0.004	0.665 $\pm$ 0.005	0.408 $\pm$ 0.015	0.181 $\pm$ 0.004	0.148 $\pm$ 0.001
PARTICLE	0.484 $\pm$ 0.015	-	0.196 $\pm$ 0.018	-	-	0.289 $\pm$ 0.043	-	-
Hamming Loss $\downarrow$								
REHC-PML	0.017 $\pm$ 0.000	0.013$\pm$0.000	0.034$\pm$0.000	0.032$\pm$0.001	0.024$\pm$0.000	0.052 $\pm$ 0.001	0.045$\pm$0.001	0.025 $\pm$ 0.000
PML-FSMIR	0.017 $\pm$ 0.000	-	0.040 $\pm$ 0.000	-	0.028 $\pm$ 0.000	0.055 $\pm$ 0.001	-	-
WML3CP	0.019 $\pm$ 0.001	0.019 $\pm$ 0.005	0.037 $\pm$ 0.001	0.040 $\pm$ 0.001	0.028 $\pm$ 0.000	0.059 $\pm$ 0.001	0.058 $\pm$ 0.003	0.025 $\pm$ 0.005
PLAIN	0.022 $\pm$ 0.000	0.016 $\pm$ 0.000	0.037 $\pm$ 0.000	0.035 $\pm$ 0.001	0.029 $\pm$ 0.000	0.055 $\pm$ 0.002	0.052 $\pm$ 0.001	0.025 $\pm$ 0.000
PARD	0.013$\pm$0.000	0.009 $\pm$ 0.000	0.035 $\pm$ 0.000	0.038 $\pm$ 0.000	0.019 $\pm$ 0.000	0.048$\pm$0.002	0.059 $\pm$ 0.000	0.023$\pm$0.000
PAMB	0.025 $\pm$ 0.000	-	0.050 $\pm$ 0.002	-	-	0.068 $\pm$ 0.008	-	-
GLC	0.023 $\pm$ 0.001	0.014 $\pm$ 0.000	0.041 $\pm$ 0.000	0.040 $\pm$ 0.001	0.037 $\pm$ 0.001	0.062 $\pm$ 0.001	0.073 $\pm$ 0.001	0.027 $\pm$ 0.000
PMLMD	0.017 $\pm$ 0.000	0.015 $\pm$ 0.000	0.039 $\pm$ 0.000	0.035 $\pm$ 0.001	0.025 $\pm$ 0.000	0.053 $\pm$ 0.001	0.052 $\pm$ 0.001	0.027 $\pm$ 0.001
PMLNI	0.016 $\pm$ 0.000	0.013$\pm$0.000	0.037 $\pm$ 0.000	0.038 $\pm$ 0.000	0.024$\pm$0.000	0.070 $\pm$ 0.001	0.061 $\pm$ 0.001	0.025 $\pm$ 0.000
PARTICLE	0.024 $\pm$ 0.000	-	0.039 $\pm$ 0.002	-	-	0.065 $\pm$ 0.009	-	-
One Error $\downarrow$								
REHC-PML	0.414$\pm$0.008	0.523$\pm$0.004	0.146$\pm$0.005	0.275$\pm$0.007	0.304$\pm$0.007	0.219$\pm$0.033	0.262$\pm$0.007	0.560$\pm$0.002
PML-FSMIR	0.432 $\pm$ 0.018	-	0.202 $\pm$ 0.008	-	0.432 $\pm$ 0.014	0.264 $\pm$ 0.034	-	-
WML3CP	0.530 $\pm$ 0.018	0.584 $\pm$ 0.000	0.180 $\pm$ 0.002	0.358 $\pm$ 0.011	0.476 $\pm$ 0.018	0.290 $\pm$ 0.019	0.406 $\pm$ 0.013	0.599 $\pm$ 0.000
PLAIN	0.648 $\pm$ 0.015	0.757 $\pm$ 0.004	0.171 $\pm$ 0.004	0.329 $\pm$ 0.003	0.523 $\pm$ 0.019	0.255 $\pm$ 0.028	0.339 $\pm$ 0.004	0.584 $\pm$ 0.002
PARD	0.429 $\pm$ 0.030	0.608 $\pm$ 0.006	0.205 $\pm$ 0.004	0.414 $\pm$ 0.015	0.404 $\pm$ 0.010	0.227 $\pm$ 0.014	0.415 $\pm$ 0.004	0.653 $\pm$ 0.003
PAMB	0.922 $\pm$ 0.013	-	0.234 $\pm$ 0.030	-	-	0.685 $\pm$ 0.105	-	-
GLC	0.740 $\pm$ 0.014	0.624 $\pm$ 0.004	0.219 $\pm$ 0.005	0.410 $\pm$ 0.009	0.934 $\pm$ 0.004	0.300 $\pm$ 0.023	0.564 $\pm$ 0.005	0.675 $\pm$ 0.002
PMLMD	0.463 $\pm$ 0.010	0.665 $\pm$ 0.011	0.209 $\pm$ 0.017	0.321 $\pm$ 0.010	0.376 $\pm$ 0.014	0.243 $\pm$ 0.026	0.337 $\pm$ 0.009	0.642 $\pm$ 0.024
PMLNI	0.384 $\pm$ 0.014	0.548 $\pm$ 0.003	0.181 $\pm$ 0.005	0.379 $\pm$ 0.008	0.343 $\pm$ 0.007	0.401 $\pm$ 0.013	0.435 $\pm$ 0.009	0.595 $\pm$ 0.002
PARTICLE	0.749 $\pm$ 0.018	-	0.181 $\pm$ 0.017	-	-	0.339 $\pm$ 0.071	-	-
Ranking Loss $\downarrow$								
REHC-PML	0.100 $\pm$ 0.007	0.094$\pm$0.003	0.047$\pm$0.001	0.074 $\pm$ 0.003	0.107$\pm$0.002	0.089 $\pm$ 0.003	0.085$\pm$0.011	0.059$\pm$0.001
PML-FSMIR	0.110 $\pm$ 0.003	-	0.061 $\pm$ 0.001	-	0.147 $\pm$ 0.003	0.081$\pm$0.004	-	-
WML3CP	0.142 $\pm$ 0.005	0.159 $\pm$ 0.003	0.055 $\pm$ 0.005	0.084 $\pm$ 0.005	0.138 $\pm$ 0.002	0.129 $\pm$ 0.001	0.093 $\pm$ 0.018	0.071 $\pm$ 0.003
PLAIN	0.238 $\pm$ 0.015	0.228 $\pm$ 0.003	0.081 $\pm$ 0.003	0.104 $\pm$ 0.004	0.191 $\pm$ 0.004	0.096 $\pm$ 0.002	0.116 $\pm$ 0.003	0.091 $\pm$ 0.001
PARD	0.079$\pm$0.008	0.119 $\pm$ 0.009	0.061 $\pm$ 0.001	0.071$\pm$0.002	0.190 $\pm$ 0.008	0.089 $\pm$ 0.001	0.087 $\pm$ 0.003	0.090 $\pm$ 0.001
PAMB	0.432 $\pm$ 0.013	-	0.233 $\pm$ 0.014	-	-	0.226 $\pm$ 0.037	-	-
GLC	0.155 $\pm$ 0.005	0.100 $\pm$ 0.001	0.064 $\pm$ 0.001	0.076 $\pm$ 0.002	0.668 $\pm$ 0.006	0.104 $\pm$ 0.004	0.106 $\pm$ 0.002	0.088 $\pm$ 0.000
PMLMD	0.099 $\pm$ 0.005	0.135 $\pm$ 0.003	0.074 $\pm$ 0.004	0.074 $\pm$ 0.003	0.118 $\pm$ 0.002	0.084 $\pm$ 0.004	0.087 $\pm$ 0.004	0.100 $\pm$ 0.004
PMLNI	0.128 $\pm$ 0.003	0.117 $\pm$ 0.002	0.056 $\pm$ 0.002	0.118 $\pm$ 0.002	0.135 $\pm$ 0.000	0.177 $\pm$ 0.007	0.136 $\pm$ 0.004	0.079 $\pm$ 0.001
PARTICLE	0.327 $\pm$ 0.010	-	0.059 $\pm$ 0.006	-	-	0.111 $\pm$ 0.018	-	-

Table 1: Experimental results (mean $\pm$ std) on eight UCI PML datasets ($m = 100$). The best performance is shown in **boldface**.

achieves competitive Coverage Error, while REHC-PML excels on other datasets, demonstrating robustness across different data characteristics. Table 2 shows the performance under low noise conditions. As noise decreases, all methods improve, but REHC-PML maintains its leading position. Especially in terms of Average Precision, on *bibtex*, REHC-PML achieves 0.549 (tied with PMLNI), while substantially outperforming other baselines.

The results across all three noise levels ($m \in \{150, 100, 50\}$) reveal that as noise decreases, the relative performance gap between REHC-PML and baselines generally narrows on simpler datasets but remains substantial on

complex datasets with rich label structures (*e.g.*, *delicious*, *bookmarks*, *bibtex*). This observation suggests that high-order label correlations are valuable when dealing with complex multi-label scenarios, where pairwise correlations may not suffice to capture the complex label dependencies.

References

[Gao *et al.*, 2025] Wanfu Gao, Hanlin Pan, Qingqi Han, and Kunpeng Liu. Noise-resistant label reconstruction feature selection for partial multi-label learning. In *Proceedings of*

Methods	bibtex	bookmarks	bow-train	computers	delicious	enron	entertainment	nuswide
Average Precision $\uparrow$								
REHC-PML	0.549$\pm$0.009	0.505$\pm$0.002	0.767$\pm$0.004	0.764$\pm$0.007	0.406$\pm$0.003	0.698$\pm$0.016	0.786$\pm$0.004	0.675$\pm$0.001
PML-FSMIR	0.506 $\pm$ 0.009	-	0.670 $\pm$ 0.003	-	0.271 $\pm$ 0.006	0.670 $\pm$ 0.015	-	-
WPML3CP	0.418 $\pm$ 0.030	0.448 $\pm$ 0.001	0.698 $\pm$ 0.013	0.702 $\pm$ 0.010	0.261 $\pm$ 0.003	0.647 $\pm$ 0.007	0.697 $\pm$ 0.028	0.536 $\pm$ 0.001
PLAIN	0.308 $\pm$ 0.007	0.277 $\pm$ 0.002	0.709 $\pm$ 0.006	0.710 $\pm$ 0.003	0.222 $\pm$ 0.006	0.665 $\pm$ 0.017	0.725 $\pm$ 0.005	0.638 $\pm$ 0.002
PARD	0.521 $\pm$ 0.018	0.416 $\pm$ 0.004	0.677 $\pm$ 0.002	0.677 $\pm$ 0.009	0.274 $\pm$ 0.004	0.684 $\pm$ 0.010	0.699 $\pm$ 0.004	0.581 $\pm$ 0.002
PAMB	0.143 $\pm$ 0.011	-	0.520 $\pm$ 0.014	-	-	0.479 $\pm$ 0.075	-	-
GLC	0.286 $\pm$ 0.009	0.422 $\pm$ 0.001	0.652 $\pm$ 0.002	0.659 $\pm$ 0.007	0.053 $\pm$ 0.001	0.576 $\pm$ 0.009	0.611 $\pm$ 0.007	0.566 $\pm$ 0.001
PMLMD	0.526 $\pm$ 0.004	0.391 $\pm$ 0.007	0.683 $\pm$ 0.005	0.725 $\pm$ 0.007	0.348 $\pm$ 0.003	0.698 $\pm$ 0.015	0.738 $\pm$ 0.005	0.586 $\pm$ 0.011
PMLNI	0.549$\pm$0.003	0.473 $\pm$ 0.003	0.714 $\pm$ 0.003	0.674 $\pm$ 0.007	0.364 $\pm$ 0.002	0.566 $\pm$ 0.021	0.666 $\pm$ 0.003	0.637 $\pm$ 0.001
PARTICLE	0.278 $\pm$ 0.008	-	0.693 $\pm$ 0.014	-	-	0.597 $\pm$ 0.058	-	-
Coverage Error $\downarrow$								
REHC-PML	0.158 $\pm$ 0.005	0.139$\pm$0.002	0.160$\pm$0.004	0.103$\pm$0.007	0.494$\pm$0.010	0.239 $\pm$ 0.007	0.139 $\pm$ 0.006	0.113$\pm$0.001
PML-FSMIR	0.192 $\pm$ 0.005	-	0.197 $\pm$ 0.002	-	0.636 $\pm$ 0.006	0.232$\pm$0.010	-	-
WPML3CP	0.363 $\pm$ 0.017	0.126$\pm$0.004	0.174 $\pm$ 0.004	0.122 $\pm$ 0.006	0.612 $\pm$ 0.006	0.272 $\pm$ 0.009	0.129$\pm$0.012	0.157 $\pm$ 0.004
PLAIN	0.354 $\pm$ 0.021	0.313 $\pm$ 0.005	0.228 $\pm$ 0.004	0.155 $\pm$ 0.005	0.741 $\pm$ 0.007	0.271 $\pm$ 0.011	0.159 $\pm$ 0.006	0.167 $\pm$ 0.002
PARD	0.139$\pm$0.009	0.183 $\pm$ 0.009	0.201 $\pm$ 0.004	0.105$\pm$0.002	0.879 $\pm$ 0.020	0.246 $\pm$ 0.009	0.124$\pm$0.002	0.165 $\pm$ 0.001
PAMB	0.563 $\pm$ 0.006	-	0.538 $\pm$ 0.012	-	-	0.497 $\pm$ 0.101	-	-
GLC	0.240 $\pm$ 0.002	0.154 $\pm$ 0.001	0.204 $\pm$ 0.002	0.113 $\pm$ 0.004	0.951 $\pm$ 0.003	0.268 $\pm$ 0.011	0.141 $\pm$ 0.003	0.160 $\pm$ 0.001
PMLMD	0.156 $\pm$ 0.005	0.190 $\pm$ 0.003	0.238 $\pm$ 0.018	0.118 $\pm$ 0.006	0.544 $\pm$ 0.007	0.238 $\pm$ 0.010	0.125 $\pm$ 0.004	0.185 $\pm$ 0.005
PMLNI	0.227 $\pm$ 0.003	0.182 $\pm$ 0.002	0.186 $\pm$ 0.002	0.162 $\pm$ 0.004	0.646 $\pm$ 0.003	0.401 $\pm$ 0.023	0.168 $\pm$ 0.003	0.145 $\pm$ 0.000
PARTICLE	0.472 $\pm$ 0.009	-	0.198 $\pm$ 0.018	-	-	0.283 $\pm$ 0.041	-	-
Hamming Loss $\downarrow$								
REHC-PML	0.017 $\pm$ 0.000	0.013$\pm$0.000	0.033$\pm$0.000	0.032$\pm$0.001	0.024$\pm$0.000	0.052 $\pm$ 0.001	0.043$\pm$0.001	0.025 $\pm$ 0.000
PML-FSMIR	0.017 $\pm$ 0.000	-	0.040 $\pm$ 0.001	-	0.028 $\pm$ 0.000	0.055 $\pm$ 0.001	-	-
WPML3CP	0.019 $\pm$ 0.001	0.019 $\pm$ 0.006	0.036 $\pm$ 0.001	0.04 $\pm$ 0.001	0.028 $\pm$ 0.000	0.055 $\pm$ 0.000	0.057 $\pm$ 0.005	0.029 $\pm$ 0.006
PLAIN	0.022 $\pm$ 0.000	0.016 $\pm$ 0.000	0.037 $\pm$ 0.000	0.035 $\pm$ 0.001	0.029 $\pm$ 0.000	0.055 $\pm$ 0.002	0.051 $\pm$ 0.001	0.025 $\pm$ 0.000
PARD	0.013$\pm$0.000	0.009 $\pm$ 0.000	0.033$\pm$0.000	0.038 $\pm$ 0.000	0.019$\pm$0.000	0.047$\pm$0.001	0.059 $\pm$ 0.000	0.023$\pm$0.000
PAMB	0.025 $\pm$ 0.000	-	0.050 $\pm$ 0.002	-	-	0.068 $\pm$ 0.008	-	-
GLC	0.023 $\pm$ 0.000	0.014 $\pm$ 0.000	0.041 $\pm$ 0.000	0.040 $\pm$ 0.001	0.037 $\pm$ 0.001	0.062 $\pm$ 0.002	0.073 $\pm$ 0.002	0.028 $\pm$ 0.000
PMLMD	0.017 $\pm$ 0.000	0.015 $\pm$ 0.000	0.038 $\pm$ 0.001	0.035 $\pm$ 0.001	0.025 $\pm$ 0.000	0.052 $\pm$ 0.001	0.051 $\pm$ 0.001	0.027 $\pm$ 0.001
PMLNI	0.016 $\pm$ 0.000	0.013$\pm$0.000	0.037 $\pm$ 0.000	0.038 $\pm$ 0.001	0.024$\pm$0.000	0.066 $\pm$ 0.003	0.060 $\pm$ 0.001	0.025 $\pm$ 0.000
PARTICLE	0.023 $\pm$ 0.000	-	0.039 $\pm$ 0.002	-	-	0.064 $\pm$ 0.009	-	-
One Error $\downarrow$								
REHC-PML	0.402$\pm$0.011	0.518$\pm$0.003	0.144$\pm$0.005	0.277$\pm$0.005	0.304$\pm$0.010	0.232$\pm$0.038	0.250$\pm$0.004	0.559$\pm$0.002
PML-FSMIR	0.423 $\pm$ 0.019	-	0.204 $\pm$ 0.008	-	0.428 $\pm$ 0.008	0.264 $\pm$ 0.028	-	-
WPML3CP	0.558 $\pm$ 0.038	0.575 $\pm$ 0.003	0.190 $\pm$ 0.003	0.359 $\pm$ 0.01	0.489 $\pm$ 0.018	0.236 $\pm$ 0.015	0.401 $\pm$ 0.057	0.717 $\pm$ 0.003
PLAIN	0.636 $\pm$ 0.009	0.758 $\pm$ 0.003	0.170 $\pm$ 0.005	0.330 $\pm$ 0.004	0.524 $\pm$ 0.014	0.253 $\pm$ 0.022	0.333 $\pm$ 0.008	0.584 $\pm$ 0.002
PARD	0.437 $\pm$ 0.023	0.611 $\pm$ 0.006	0.205 $\pm$ 0.004	0.414 $\pm$ 0.016	0.404 $\pm$ 0.015	0.223 $\pm$ 0.013	0.405 $\pm$ 0.005	0.655 $\pm$ 0.003
PAMB	0.921 $\pm$ 0.013	-	0.212 $\pm$ 0.024	-	-	0.609 $\pm$ 0.173	-	-
GLC	0.736 $\pm$ 0.015	0.622 $\pm$ 0.002	0.219 $\pm$ 0.004	0.410 $\pm$ 0.009	0.930 $\pm$ 0.003	0.297 $\pm$ 0.016	0.564 $\pm$ 0.013	0.678 $\pm$ 0.001
PMLMD	0.445 $\pm$ 0.008	0.647 $\pm$ 0.009	0.210 $\pm$ 0.018	0.323 $\pm$ 0.011	0.374 $\pm$ 0.010	0.230 $\pm$ 0.037	0.323 $\pm$ 0.005	0.643 $\pm$ 0.025
PMLNI	0.379$\pm$0.007	0.547 $\pm$ 0.003	0.181 $\pm$ 0.005	0.373 $\pm$ 0.012	0.342 $\pm$ 0.005	0.361 $\pm$ 0.020	0.420 $\pm$ 0.008	0.596 $\pm$ 0.002
PARTICLE	0.737 $\pm$ 0.017	-	0.180 $\pm$ 0.019	-	-	0.326 $\pm$ 0.058	-	-
Ranking Loss $\downarrow$								
REHC-PML	0.091 $\pm$ 0.004	0.089$\pm$0.002	0.044$\pm$0.001	0.070$\pm$0.006	0.104$\pm$0.002	0.080$\pm$0.003	0.092$\pm$0.004	0.058$\pm$0.001
PML-FSMIR	0.108 $\pm$ 0.003	-	0.060 $\pm$ 0.001	-	0.146 $\pm$ 0.003	0.081$\pm$0.004	-	-
WPML3CP	0.201 $\pm$ 0.022	0.138 $\pm$ 0.004	0.051 $\pm$ 0.005	0.079 $\pm$ 0.004	0.138 $\pm$ 0.002	0.099 $\pm$ 0.005	0.092 $\pm$ 0.013	0.091 $\pm$ 0.004
PLAIN	0.232 $\pm$ 0.013	0.224 $\pm$ 0.004	0.070 $\pm$ 0.002	0.105 $\pm$ 0.004	0.190 $\pm$ 0.004	0.096 $\pm$ 0.005	0.112 $\pm$ 0.004	0.089 $\pm$ 0.001
PARD	0.078$\pm$0.007	0.117 $\pm$ 0.004	0.060 $\pm$ 0.001	0.068$\pm$0.002	0.216 $\pm$ 0.014	0.086 $\pm$ 0.004	0.087 $\pm$ 0.001	0.090 $\pm$ 0.001
PAMB	0.430 $\pm$ 0.006	-	0.215 $\pm$ 0.010	-	-	0.227 $\pm$ 0.045	-	-
GLC	0.155 $\pm$ 0.003	0.098 $\pm$ 0.000	0.064 $\pm$ 0.001	0.075 $\pm$ 0.002	0.676 $\pm$ 0.004	0.105 $\pm$ 0.004	0.107 $\pm$ 0.003	0.088 $\pm$ 0.001
PMLMD	0.084 $\pm$ 0.004	0.122 $\pm$ 0.003	0.070 $\pm$ 0.003	0.075 $\pm$ 0.003	0.111 $\pm$ 0.001	0.080$\pm$0.004	0.085$\pm$0.004	0.101 $\pm$ 0.005
PMLNI	0.123 $\pm$ 0.003	0.113 $\pm$ 0.002	0.054 $\pm$ 0.001	0.112 $\pm$ 0.003	0.129 $\pm$ 0.001	0.168 $\pm$ 0.008	0.124 $\pm$ 0.002	0.077 $\pm$ 0.001
PARTICLE	0.316 $\pm$ 0.006	-	0.060 $\pm$ 0.006	-	-	0.108 $\pm$ 0.018	-	-

Table 2: Experimental results (mean $\pm$ std) on eight UCI PML datasets ($m = 50$). The best performance is shown in **boldface**.

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